- By moving the jockey over the wire, the point on the wire at which the galvanometer shows null deflection i.e., balancing point $J$ is found.
- The balancing length $AJ = l$ is noted.
- The unknown resistance $X_1$ is found using the formula $X_1 = \frac{R(100-l)}{l}$.
- The experiment is repeated for different values of $R$.
- The same procedure is repeated after interchanging $R$ and $X$.
- The unknown resistance $X_2$ is found using the formula $X_2 = \frac{Rl}{(100-l)}$.
- The experiment is repeated for same values of $R$ as before.
- The resistance of the given coil is found from the mean value of $X_1$ and $X_2$.
- The radius of the wire $r$ is found using screw gauge.
- The length of the coil $L$ is measured using meter scale.
- From the values of $X$, $r$ and $L$, the specific resistance of the material of the wire is determined.

## OBSERVATION

length of the coil, $L$ = _____ cm.

Table 1 To find the resistance of the given coil

| S.No. | Resistance R (Ω) | Before interchanging | | After interchanging | | Mean $X = \frac{X_1 + X_2}{2}$ (Ω) |
|---|---|---|---|---|---|---|
| | | Balancing length $l$ (cm) | $X_1 = \frac{R(100-l)}{l}$ (Ω) | Balancing length $l$ (cm) | $X_2 = \frac{Rl}{(100-l)}$ (Ω) | |
| 1 | | | | | | |
| 2 | | | | | | |
| 3 | | | | | | |

Mean resistance, X = ----------Ω



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